

A Generalized Nash Equilibrium-Seeking Scheme for Trauma Resuscitation

Promise Ekpo* Angelique Taylor* Lekan Molu**

* Cornell University, NY. {poe6, amt298}@cornell.edu.

** Bala Cynwyd, PA. lekanmolu@scriptedonachip.com

Abstract: Trauma resuscitation is a clinical process for treating life-threatening physiological disorders in safety-critical environments, driven by the experience of the healthcare workers (HCW) team. Designing and optimizing quantifiable metrics that accurately capture HCW decisions may augment current resuscitation procedures with the potential to improve patient outcomes. This motivates our socio-technical formulation of trauma resuscitation as a distributed generalized Nash equilibrium (GNE)-seeking game with coupled inequality constraints. This method is optimized over a time-varying communication graph. We introduce novel insights from clinical experience to model HCWs behavior. This work informs group dynamics equilibrium that constitutes the best possible resuscitation outcome given HCWs' workloads, schedules, and competencies, resource allocation, and systemic constraints.

Keywords: Cyber-physical and human systems (CPHS); Social computing; Game theories.

1. INTRODUCTION

Trauma resuscitation (TR) is a collaborative, often conflicting, coordination effort among a team of healthcare workers (HCWs). It is often characterized by significant differential in (i) skillset; (ii) mental and physical alacrity; and (iii) intra-team communication skills. These factors are shaped by HCWs' workplace experience and can degrade over time. The degradation may spur burnout and fatigue — undermining productivity; miscommunication informs duplicated execution of tasks — engendering redundancy. Across organization verticals to peer horizontals in the medical environment, through verbal and aural cues (Sarcevic et al., 2012) HCWs can improve their decision quality. Their decisions can be improved if a systematic and quantifiable means exist to integrate these cues into an algorithmic framework whose performance can be consistently tracked. Previous efforts have integrated autonomous systems to support team dynamics (Taylor et al., 2024), deliver supplies (Salinas-Avila et al., 2022), or sanitize surgical rooms (Sun and Yeung, 2007). Our recent effort addressed equitable workload allocation in a multi-agent setting by studying the problem in our controlled MARLHospital simulator (Ekpo et al., 2025a).

A well-executed TR procedure involves a deliberative process where skill-informed task delegation by clinic leaders and responsibilities hand-off among peers are executed efficiently despite time constraints. Which type of autonomous decision-making process can we devise for better outcomes in rapidly changing, high-stakes safety-critical socio-technical environments? Such a process should (i) support clinical decision-making through automated task allocation among HCWs; (ii) maximize HCW skill levels while balancing other HCWs' skill levels; and (iii) enhance communication efficacy while reducing duplicate information exchange. In this paper,

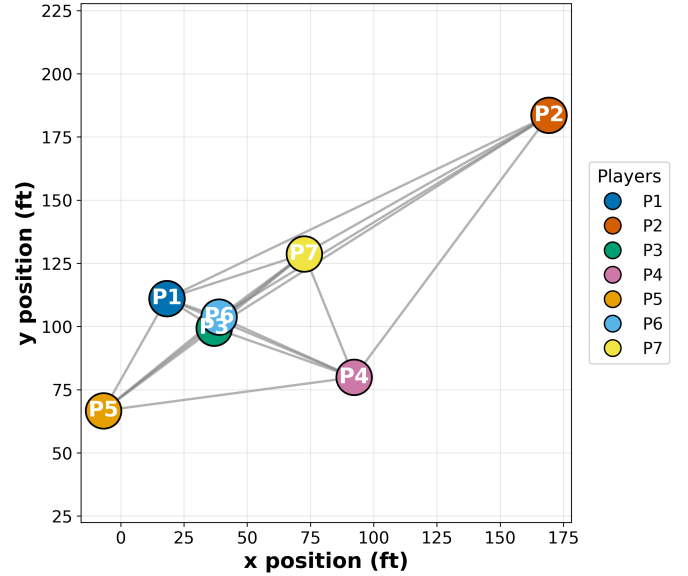


Fig. 1. Communication graph $\mathcal{G}(\mathcal{A})(k)$ at $k = 199$. Nodes are $n = 7$ healthcare workers; edges link pairs within radius $r = 200$ ft. Player positions yield a time-varying neighbor structure.

we study quantifiable algorithms in socio-technical, high-stakes medical settings to enhance patient outcomes such that as TR proceeds, individual HCWs examine their available objectives and decide optimization variables.

We resort to a game-theoretic decision-making mechanism in collaborative TR settings, informed by our observations in a representative medical simulation training environment for HCWs at Weill Cornell Emergency Medicine in

New York City¹. We provide a mathematical characterization between observed skill levels and task coordination models, to deliver an algorithmic framework to optimize TR procedures. The rest of this paper is structured as follows: §2 describes the notations and games background. Our formulation is presented in §3 and numerical evaluations are provided in §4. We conclude the paper in §5.

2. BACKGROUND

2.1 Multiagent Healthcare Automation

The role of shared mental models and decentralized communication workflows has been described in high-duress trauma resuscitation contexts (Sarcevic et al., 2012; Batra et al., 2026). Co-design studies have proposed cognitive aids and augmented-reality systems to improve information flow and situational awareness for distributed clinical teams (Taylor et al., 2024). MATEC (Cho et al., 2025) and TraumaFlow (Neumann et al., 2024) demonstrate that structured software agents can align team actions with clinical protocols even with centralized controllers.

Distributed optimization in games over networks can lead to a generalized Nash equilibrium (GNE), which was proposed for multi-agent social agents with coupling constraints in (Dave et al., 2022). Models incorporating agent dynamics, constraints, and communication topology align well with real-world healthcare teams; here, agents exhibit bounded rationality, partial observability, and asymmetric influence.

2.2 Trauma Resuscitation as a Distributed Game

How do we effectively model subjective human decisions, recommend actions, and provide failure-proof decisions for patient resuscitation? Critically, since the feasible action set of each HCW (or player) in the ER (game) is conditioned on those of other players, a generalized Nash equilibrium (GNE) solution concept is amenable to our problem. We model TR as a computer-supported cooperative work with critical information from player decisions that are relevant to the patient’s outcome. As each player does not possess full observability of every other player’s action in the system, we cast the game in a partially observable setting where the players are *noncooperative* (that is each player acts selfishly to extremize its given objectives). However, collaborative information sharing with neighbors minimizes the fuzziness of global information awareness. As the TR process proceeds in practice, individual HCWs examine their locally available pre-assigned objectives and determine which objective to optimize. Thus, searching for a GNE is equivalent to finding the phase where every player’s action are optimal with respect to their own objective and no single player can improve their objective by unilaterally changing their action.

2.3 General Notations

The set of real (non-negative) numbers is denoted, $\mathbb{R}$ ($\mathbb{R}_+$), the m -dimensional vector space is $\mathbb{R}^m$, and the $n \times n$

¹ The BASE Camp event for interprofessional pediatric emergency medicine team training in the emergency care of critically ill and injured children was employed (Weill Cornell Medicine, 2025).

dimensional real matrix is denoted $\mathbb{R}^{n \times n}$. A time-varying variable, x is denoted $x(t)$. For a function f that depends on $x(t)$, we write $f(x;t)$. The absolute value of scalar x is $|x|$. The Euclidean norm of the vector $x \in \mathbb{R}^n$ is $\|x\| := \sqrt{x^\top x}$. The n -dimensional vector of ones (zeroes) is 1_n (0_n). For a differentiable function $J(x)$, the first-order derivative of $J(x)$ is denoted $\nabla_x J(x)$. For a set Ω and variable x , the Euclidean projection of x onto Ω is $\mathcal{P}_\Omega(x) = \operatorname{argmin}_{x' \in \Omega} \|x - x'\|_2$. The set $\mathcal{S} \subset \mathbb{R}^m$ is a convex set if for $\alpha > 0$ and for every $x, y \in \mathcal{S}$, $\alpha x + (1 - \alpha)y \in \mathcal{S}$. The cardinality of a $\mathcal{S}$ is denoted $|\mathcal{S}|$. Suppose that $\mathcal{S}$ is a closed convex set, and $\mathcal{P}_\mathcal{S}(x)$ is a projection of x onto the set $\mathcal{S}$, then there is a unique element $\mathcal{P}_\mathcal{S}(x) \in \mathcal{S}$ such that $\|x - \mathcal{P}_\mathcal{S}(x)\| = \inf_{y \in \mathcal{S}} \|x - y\|$. Similarly, the Euclidean projection of x onto the set Ω is denoted $\mathcal{P}_\Omega(x) = \operatorname{argmin}_{y \in \Omega} \|x - y\|$.

2.4 Games Taxonomy

Let a game be denoted as $\Gamma(\mathcal{V}, \Omega, J)$, where $\mathcal{V}$ are the players $\{1, \dots, n\}$, each with action x_i belonging in a constraint set $\Omega_i \in \mathbb{R}^m$ and a local cost profile $J_i : \Omega_i \rightarrow \mathbb{R}$ so that $J = \{J_1, \dots, J_n\}$. The action profile of the game is $\Omega := \Omega_1 \times \dots \times \Omega_n$. The vector formed by all players’ actions is $x := (x_i, x_{-i}) \triangleq \{x_i\}_{i=1}^n$, where the vector formed by all the players’ actions except those of player i is $x_{-i} = \{x_{i'}\}_{i'=1, i' \neq i}^n \in \mathbb{R}^{n-i}$ where $n_{-i} := n - n_i$.

Define $K_i^p := \{x_i \in \Omega_i \mid h_i(x_i) \leq 0\}$ as the set of each individual player’s constraint set, and $K^s := \{x \in \Omega \mid g(x) \geq 0_m\}$ as the set of shared constraints, where $g(x) = c^\top x - d$, for $c = [c_1^\top, \dots, c_n^\top]^\top$, $c_i \in \mathbb{R}^m$ and $d \in \mathbb{R}$. It follows that player i ’s action is constrained along two directions: constraints that depend on the action of other players i.e. $g(x) \geq 0$, and individual constraints that depend on player i ’s action i.e. $x_i \in K_i$. Contrary to popular formulations in GNE-seeking literature, we have chosen $g(x) \geq 0$ since the skills, time indices, and other parameters we are optimizing for cannot be negative.

2.5 Games Machinery

Let us set a few definitions and preliminary results in motion. Denote the feasible set of player i ’ action by $K_i(x_{-i}) := \{x_i \mid (x_i, x_{-i}) \in K\}$, where $K := K^s \cap (K_1 \times \dots \times K_n)$. Suppose we define the generalized Nash equilibrium problem (GNEP) as $\Gamma(\mathcal{V}, \Omega, J, K)$, then the individual players in the game “freeze” other players’ actions x_{-i} as exogenous variables, to solve

$$\begin{aligned} & \min_{x_i \in \Omega_i} J_i(x_i, x_{-i}), \quad \forall i \in \mathcal{V} \\ & \text{subject to } x_i \in K_i(x_{-i}). \end{aligned} \quad (\text{GNEP})$$

Definition 1. (Neighbors of a Player). We define the neighbors $\mathcal{N}_i(t)$ of player i at time t as the set of all agents that lie within a predefined radius, r_i . $\mathcal{N}_i(t) = \{j \mid (j, i) \in \mathcal{E}\}$.

Definition 2. (Generalized Nash Equilibrium). In problem (GNEP), the action profile $x^* := (x_i^*, x_{-i}^*)$ is a GNE if $J_i(x_i^*, x_{-i}^*) \leq J_i(x_i, x_{-i}^*), \forall x_i \in \Omega_i, \forall i \in \mathcal{V}$.

Assumptions are made for feasibility of the solution (GNEP).

Assumption 1. (Interconnectivity). The game occurs over a time-varying *connected*² and undirected³ weighted communication graph $\mathcal{G}(\mathcal{A})(t) : \{\mathcal{V}, \mathcal{E}\}$ where $t > 0$ with edges $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ and interconnections' adjacency matrix $\mathcal{A} := [a_{ij}] \in \mathbb{R}^{n \times n}$. Player i is connected to (can receive information from) player j if $(i, j) \in \mathcal{E}$ and vice versa. Let $a_{ij} > 0$ if $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. For a $D = \text{diag}(d_{11}, \dots, d_{nn}) \in \mathbb{R}^{n \times n}$, where $d_{ii} = \sum_{j=1}^n a_{ij}$ for all $i \in \mathcal{V}$, we have $\mathcal{G}(\mathcal{A})$'s Laplacian as $\mathcal{L} := D - \mathcal{A} \in \mathbb{R}^{n \times n}$

Assumption 2. (Cost Function). For $i \in \mathcal{V}$, $\Omega_i \in \mathbb{R}^m$ is a nonempty, convex, and closed set; and the cost $J_i(x_i, x_{-i})$ is continuously differentiable on Ω and convex in x_i for every fixed x_{-i} . Define $\nabla J(x) = [\nabla_{x_1} J_1^\top(x), \dots, \nabla_{x_n} J_n^\top(x)]^\top$ as the *single-value* game mapping.

Assumption 3. (Monotonicity and Game-mapping). The single-valued mapping $\mathcal{M} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is strongly monotone on the action constraint set $\Omega \in \mathbb{R}^m$ if there exists a constant $\mu > 0$ for any $x, x' \in \Omega$, where $x \neq x'$ such that $(\nabla J - \nabla J')^\top (x - x') \geq \mu \|x - x'\|^2$ for any $\nabla J \in \mathcal{M}(x)$, $\nabla J' \in \mathcal{M}(x')$.

Assumption 4. (Constraint Function). The inequality constraint function $g(x) \geq 0$ is continuously differentiable and convex in x . Also, the feasible action set K is nonempty, convex, and closed — satisfying the Slater's condition *i.e.*, $x^* \in \text{int}(\Omega)$ ⁴ such that $g(x^*) \leq 0_m$.

Lemma 1. (Existence of a GNEP). Let the GNEP of Def. 2 be given and suppose that (i) $\Omega_i \in \mathbb{R}^m$ be a nonempty, convex, and compact such that for $i \in \mathcal{V}$, $K_i(x_{-i})$ is nonempty, convex, closed, and K_i is both upper and lower semi-continuous; and (ii) the local cost $J_i(x_i, x_{-i})$ is quasi-convex on $K_i(x_{-i})$ for every $i \in \mathcal{V}$. Then, a GNE exists.

Proof. (Existence of GNEP) Follows directly from Thm. 4.1 in (Facchinei and Kanzow, 2010).

Remark 1. When assumptions 2, 3, and 4 are fulfilled, then the GNEP $\Gamma(\mathcal{V}, \Omega, J, K)$ satisfies the existence condition x^* in Lemma 1.

Lemma 2. (Optimality of the GNE). Suppose that there exists a multiplier $\lambda_i^* \in \mathbb{R}^m$, $\forall i \in \mathcal{V}$. The optimality condition for each player can be found via the KKT conditions

$$\begin{aligned} \nabla_{x_i} J_i(x_i^*, x_{-i}^*) + \langle \lambda_i^*, \nabla_{x_i} g(x^*) \rangle &= 0_n, \quad \forall i \in \mathcal{V}, \lambda_i^* \in \mathcal{V}, \\ 0_m &\leq \lambda_i^* \perp g(x_i^*, x_{-i}^*) \geq 0_m \end{aligned} \quad (1)$$

under the assumption of continuous differentiability. This follows from Thm. 4.6 in Facchinei and Kanzow (2010).

As a system of equations, (1) can be written as

$$\begin{aligned} J(x^*, \lambda^*) &= 0, \\ 0 &\leq \lambda^* \perp g(x^*) \geq 0, \end{aligned} \quad (2)$$

for $g(x^*) = [g^\top(x_1^*), \dots, g^\top(x_n^*)]^\top$, $\lambda = [\lambda_1^\top, \dots, \lambda_n^\top]^\top$, and $J(x^*, \lambda^*) = [\nabla_{x_1} J_1^\top(x_1^*), \dots, \nabla_{x_n} J_n^\top(x_n^*)]^\top$.

With the constraint qualification, the x portion of (2) is a first-order necessary condition for the GNEP; and under

Table 1. Representative HCW skill attributes (ALS protocol).

Taxonomy	Meaning
CPR	Cardiopulmonary resuscitation
SHOCK	Deliver defibrillation shock
RHYTHM_CHECK	Evaluate rhythm every 2 min
AIRWAY_MGMT	Airway interventions (OPA/NPA/ETT)
EPI	Epinephrine administration
TEAM_COMM	Critical team communication

appropriate convexity assumptions, the x part solves the GNEP so that (2) is sufficient condition for the GNEP.

Remark 2. Lemma 2 is the so-called variational inequality problem $\text{VI}(X, \nabla J(x))$ whose solution set is a special class of GNEs with all Lagrange multipliers equal. That is, the solution x^* to the $\text{VI}(X, \nabla J(x))$ under the KKT conditions (1) satisfies the GNE x^* if $\lambda_1^* = \dots = \lambda_n^* = \lambda^*$.

3. A GNEP TRAUMA RESUSCITATION SCHEME

We now translate clinical attributes (skill proficiency, alacrity, energy, communication efficacy) into mathematical quantities determining each worker's cost, feasible actions, and interaction structure.

3.1 Clinical model

Each healthcare worker is a player $i \in \mathcal{V}$ with objective J_i (e.g., start CPR), individual constraints K_i^p (e.g., hemorrhage control), and shared constraints K_i^s (e.g., breathing or pulse checks) in a game $\Gamma(\mathcal{V}, \Omega, \{J_i\}, K)$ where $K_i = K_i^s \cap K_i^p$. Table 1 itemizes typical HCW skill attributes inspired by the ALS code card for adult cardiac arrest (American Red Cross, 2025). The inequality constraint $h_i(x_i) \leq 0$ represents each player's clinical role assignment.

The state x_i of each player is

$$x_i(t) := [s_i, a_i(t), f_i(t), v_i(t)]^\top \quad (3)$$

with skill proficiency index s_i and time-dependent decision variables: alacrity index $a_i(t)$, fairness index $f_i(t)$, and communication efficiency $v_i(t)$. Here s_i is a fixed parameter characterizing player i 's competence, while $[a_i(t), f_i(t), v_i(t)]$ constitutes player i 's action and the state of the dynamical system that converges to the Nash equilibrium. The action profile lies in $\Omega = K_1 \times \dots \times K_n \subset \mathbb{R}^{3n}$.

The skill proficiency index $s_i \in [0, 1]$ is a fixed scalar averaging player i 's competence over past assignments and training (nursing, clinical certifications, physician training), calibrated by the institution. Performance degradation due to fatigue is captured indirectly through the time-varying alacrity index $a_i(t)$.

Players' transient step response is described as a second-order system (Nise, 2019)

$$\begin{aligned} \sigma_i(t) &= 1 - \frac{1}{\sqrt{1 - \zeta^2}} \exp(-\zeta \omega_n t) \cos(\omega_n \sqrt{1 - \zeta^2} t - \phi) \end{aligned} \quad (4)$$

where $\phi = \arctan(\zeta / (\sqrt{1 - \zeta^2}))$ for damping ratio ζ and natural frequency ω_n . We use an underdamped model

² That is, any two nodes in $\mathcal{V}$ are connected by a path.

³ That is, $(i, j) \in \mathcal{E} \Leftrightarrow (j, i) \in \mathcal{E}$.

⁴ The relative interior of the set Ω is denoted $\text{int}(\Omega)$.

($\zeta < 1$) since human physiological responses to stress overshoot before settling (Rosenblum et al., 2025). We adopt $\omega_n^i := 2\pi b_i/60$ where b_i is player i 's resting heart rate (bpm) — a measurable proxy for the engagement transient time-scale, not a control-theoretic identity⁵.

Alacrity measure: The alacrity index a_i encodes a player's readiness to execute tasks within their skillset. We model a_i via the rise time t_r of (4) — the time for $\sigma(t)$ to go from 10% to 90% of its final value. Fixing $\zeta = k < 1$ and using $\omega_n^i t$ as the normalized time variable, $a_i(t)$ is found as

$$a_i(t) = \Delta\sigma(t)/\omega_n^i, \quad t \geq 0. \quad (5)$$

Fairness in workload distribution should be encouraged when players share resources. If one worker bears workload beyond their capacity, the uneven throughput becomes untenable for efficient resuscitation. We adopt the Jain fairness index (Jain et al., 1984; Ekpo et al., 2025b) — convex, scale-invariant, population-independent, and bounded in $[1/n, 1]$:

$$f_i(w; t) = \frac{(\sum_{i=1}^n w(t))^2}{n \sum_{i=1}^n w(t)^2}, \quad w(t), t \geq 0 \quad (6)$$

which measures the equity of work w allocated to player i . Each player maintains a local team-level estimate $\hat{f}_i(t)$ obtained by running dynamic averaging consensus on $\sum_j w_j$ and $\sum_j w_j^2$. The per-task weights $\delta_\tau \in [0, 1]$ that define the workload $w_i(t) = \sum_{\tau \in \mathcal{T}_i(t)} \delta_\tau$ are specified in Section 4.

Communication efficiency. We define the communication efficacy as a function of (i) settling time; (ii) engagement level; and (iii) response time. The **settling time** t_s^i is the time for $\sigma_i(t)$'s damped oscillations to reach and stay within $\pm 2\%$ of steady state, approximated as

$$t_s^i = -\ln(\epsilon\sqrt{1-\zeta^2})/\zeta\omega_n \quad (7)$$

with $\epsilon = 0.02$. The **response time** is the difference between reaching steady state and being within 10% of $\sigma_i(t)$'s final value upon a workload handover. The timely response is $\bar{t}_i = k\Delta t_{ij}$ where Δt_{ij} is the inter-call time between players i and j for all $j \in x_{-i}$. The **engagement level** for player i is a running mean of visit counts from neighbors $\mathcal{N}_i$ up to discretized time $\mathcal{K}$:

$$e_i(t) = \frac{1}{Tn_{-i}} \int_{t_0}^T \left(\sum_{k=1}^{\mathcal{K}} \nu_{ij}(t) \right) dt, \quad \forall j \in x_{-i}.$$

A "visit" is a discrete communication event (verbal hand-off, task delegation, closed-loop confirmation) recorded at each timestep k . The **communication efficiency index** is

$$v_i(t) = \alpha(t_s^i + \bar{t}_i) + (1 - \alpha)e_i(t), \quad \alpha \in (0, 1). \quad (8)$$

3.2 Local costs and local aggregative dynamics

Problem 1. We want a distributed algorithm for a GNEP $\Gamma(\mathcal{V}, \Omega, \nabla J(x), K)$ for the socio-technical aggregative game with coupled constraints, K .

⁵ In practice, sensor noise in b_i would be handled by standard filtering (e.g., multisensor fusion, low-pass smoothing), and the projection P_{Ω_i} in (11) confines states to feasible sets regardless.

The local cost for player i is the quadratic objective between player i and its neighbors $\mathcal{N}_i$:

$$J_i(x_i, x_{-i}; t) := J_i(x_i; t) + \langle J_i(x_{-i}; t) \rangle_{r_i} \quad (9)$$

where

$$J_i(x_i; t) = \|s_i\|^2 + \|a_i(t)\|^2 + \|f_i(w; t)\|^2 + \|v_i(t)\|^2$$

and

$$\langle J_i(x_{-i}; t) \rangle_{r_i} = \frac{1}{1 + n_i(t)} \left(J_i(x_i; t) + \sum_{j \in \mathcal{N}_i(t)} J_i(x_{-i}; t) \right) \quad (10)$$

denotes the cost of neighboring players within radius $r_i > 0$ of player i . The resuscitation coordination is modeled as a dynamic game whose actions evolve under the distributed primal-dual dynamics in (11), converging to a variational generalized Nash equilibrium (v-GNE) $x^* = (x_1^*, \dots, x_n^*)$:

$$\dot{x}_i = P_{\Omega_i} \left(x_i - \alpha \nabla_{x_i} J_i(x) + \frac{\alpha\gamma}{n} \nabla_{x_i} g_i(x_i)^\top \lambda_i \right) - x_i \quad (11a)$$

$$\dot{\lambda}_i = P_{\mathbb{R}_+^p} \left[\kappa \sum_{j \in \mathcal{N}_i} \text{sgn}(\lambda_j - \lambda_i) - g_i(x_i) \right] \quad (11b)$$

$$\dot{\zeta}_i = \rho \sum_{j \in \mathcal{N}_i} \text{sgn}(\eta_j - \eta_i), \quad \dot{\eta}_i = \zeta_i + J_i(x_{-i}) \quad (11c)$$

where $\alpha > 0$ is the primal step size; $\gamma > 0$ scales the constraint coupling, normalized by n ; and $\kappa, \rho > 0$ are the consensus gains for the dual and aggregation channels. The Lagrange multiplier $\lambda_i \in \mathbb{R}_+^p$ carries one component per inequality constraint in $g_i(x_i)$; ζ_i and η_i are auxiliary scalars implementing finite-time dynamic averaging consensus on $J_i(x_{-i})$. Initial conditions are $x_i(0) \in \Omega_i$, $\lambda_i(0) \in \mathbb{R}_+^p$, and $\zeta_i(0) = 0$. Solutions are interpreted in the Filippov sense (Cortés, 2008). Algorithm (11) extends Liang et al. (2017) to inequality constraints $g(x) \geq 0$. The constants are chosen as $\kappa > (n-1)h_1$ and $\rho > \gamma(n-1)h_2$, where

$$h_1 = \sup_{i \in \mathcal{V}} \left(\sup_{x_i \in \Omega_i} \|\nabla_{x_i} J_i(\cdot, x_{-i})\| \sup_{y, y' \in \Omega} \|y - y'\| \right),$$

$$h_2 = \sup_{i \in \mathcal{V}} \left(\sup_{x_i \in \Omega_i} \|g_i(x_i)\| \right),$$

computed distributively as in (Liang et al., 2017).

4. NUMERICAL RESULTS

Setup. We validate the distributed GNEP algorithm on a trauma resuscitation scenario with $n = 7$ healthcare workers executing the Advanced Life Support (ALS) cardiac arrest protocol (American Red Cross, 2025). The synthetic dataset models a 20-minute episode with clinically accurate CPR cycles, rhythm checks, shock delivery, drug administration, airway management, and ROSC assessment. Each timestamp assigns one atomic action per agent (e.g., `Start_CPR`, `Deliver_Shock`, `Administer_Epinephrine`), following the ideal ALS flowchart.

Initial conditions and constraints. Workers are initialized with heterogeneous state components: alacrity in $[0.5, 1.0]$, fairness in $[0.8, 0.95]$, communication efficiency v_i in $[0.9, 1.0]$, and fixed skill proficiencies in $[0.6, 0.9]$. A spatial communication radius $r = 200$ ft induces time-varying neighbor sets $\mathcal{N}_i(t)$. The shared constraint $g(x) \geq$

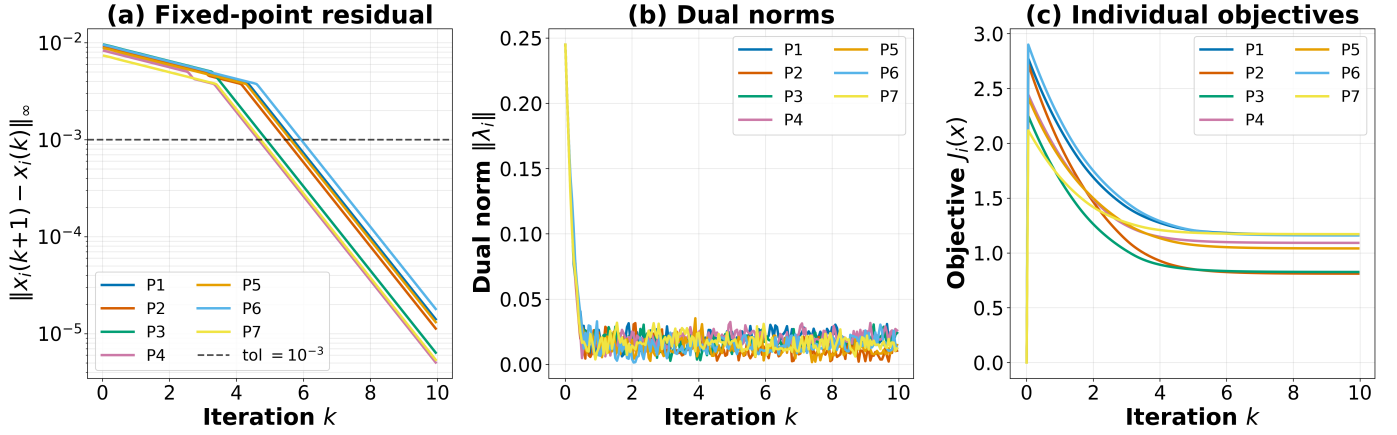


Fig. 2. Distributed GNEP simulation, $n = 7$ players. (a) Fixed-point residual. (b) Dual variable norms. (c) Individual objectives.

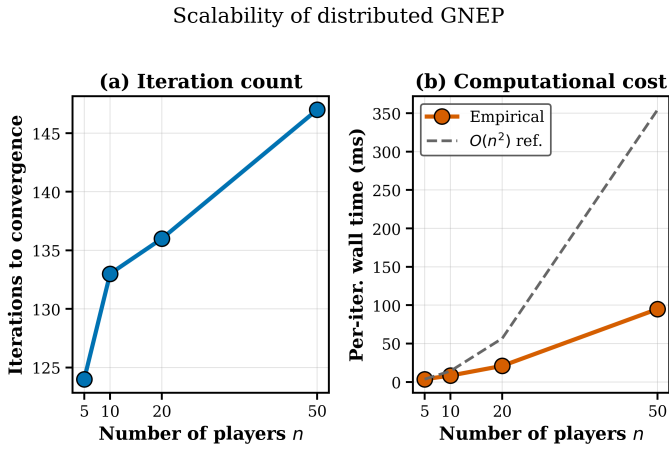


Fig. 3. Scalability of the distributed GNEP algorithm. (a) Iteration count to reach the convergence threshold 10^{-3} versus n . (b) Wall-clock time versus n , with $O(n^2)$ reference (dashed).

0 enforces minimum operational thresholds $a_i, f_i, v_i \geq 0.2$, preventing excessive fatigue or coordination degradation. All states are normalized to $[0, 1]$.

Task weighting. Each task τ is assigned a weight $\delta_\tau \in [0, 1]$, with negative penalties applied when a player's assignment falls below the protocol-prescribed threshold: -1.0 for suboptimal role assignment, -0.5 for low alacrity, and -0.5 for low energy. The negative values discount the workload contribution of substandard assignments so that the resulting workload $w_i(t) = \sum_{\tau \in \mathcal{T}_i(t)} \delta_\tau$ enters the Jain fairness index in (6) with the right sign.

Convergence. Figure 2 shows the simulation results. Fixed-point residuals decay below the 10^{-3} tolerance within roughly six iterations (panel a). Dual norms remain bounded after the initial transient (panel b). Individual objectives reach heterogeneous steady values consistent with the KKT conditions of a v-GNE (panel c).

Baseline. The baseline is a centralized greedy task-assignment scheme that assigns each pending ALS atomic action to the player with the highest remaining $s_i \cdot a_i(t)$ score. It ignores coupled constraints.

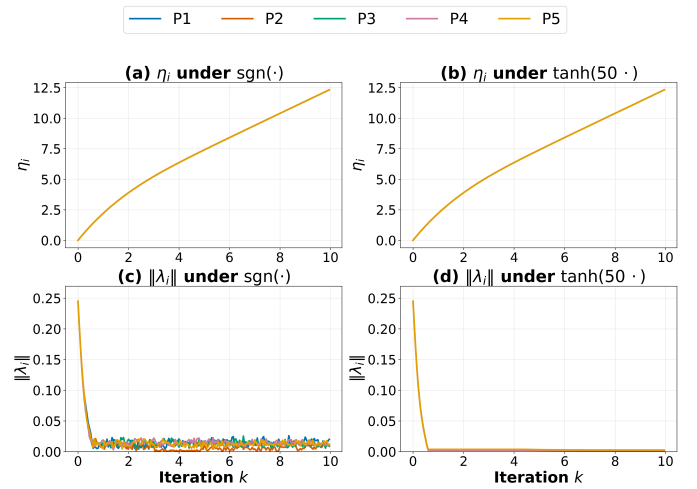


Fig. 4. Discontinuous vs. smoothed signum, $n = 5$ players. (a, b) Aggregation variable η_i under $\text{sgn}(\cdot)$ and $\tanh(50 \cdot)$. (c, d) Dual norms $\|\lambda_i\|$ under each.

Scalability. Figure 3 shows iteration count and cost as n grows. Each player's cost is dominated by the sum of its neighbors' costs, of the form $O(n + |\mathcal{E}|)$. Iteration count depends on the algebraic connectivity $\lambda_2(L)$ of the Laplacian, which is bounded away from zero for connected graphs. Empirically, iterations grow from ~ 124 at $n = 5$ to ~ 147 at $n = 50$, and wall-clock time tracks the $O(n^2)$ reference.

Chattering analysis. To mitigate chattering introduced by the signum function in (11), we smooth it with a “smooth-abs” function

$$\text{sgn}_\alpha(x) := \sqrt{x^2 + \alpha^2} - \alpha, \quad \alpha > 0, \quad (12)$$

following Tassa et al. (2012). The original discontinuous dynamics are interpreted in the standard Filippov framework of Cortés (2008). Figure 4 compares the two; smoothing preserves convergence to the v-GNE and eliminates the finite-step oscillations.

5. CONCLUSION

This paper formulated trauma resuscitation as a distributed generalized Nash equilibrium problem in which

healthcare worker skill, alacrity, fairness, and communication efficiency enter as coupled decision variables. A primal–dual seeking algorithm with inequality constraints was proposed and analyzed. Simulations on a seven-player Advanced Life Support scenario demonstrated convergence with bounded dual variables, and a sweep over up to 50 players characterized the dependence of iteration count on team size. This is a feasibility study on a synthetic clinical workflow that does not capture protocol deviations or stress-induced behavior in the decision-making process.

REFERENCES

- American Red Cross (2025). Adult Cardiac Arrest. URL <https://tinyurl.com/2jypcw6d>.
- Batra, Y., Pioldi, G., Ekpo, P., Sayatqzy, A., Maruur, P., Otieno, S., Ching, K., and Taylor, A. (2026). Rfm-hri: A multimodal dataset of medical robot failure, user reaction and recovery preferences for item retrieval tasks. doi:10.48550/arXiv.2603.05641. URL <http://arxiv.org/abs/2603.05641>. ArXiv:2603.05641 [cs].
- Cho, A., Woo, J.M., Shi, B., Udeshi, A., and Woo, J.S.H. (2025). The application of matec (multi-ai agent team care) framework in sepsis care.
- Cortés, J. (2008). Discontinuous dynamical systems: A tutorial on solutions, nonsmooth analysis, and stability. *IEEE Control Systems Magazine*, 28(3), 36–73. doi:10.1109/MCS.2008.919306.
- Dave, A., Chremos, I.V., and Malikopoulos, A.A. (2022). Social media and misleading information in a democracy: A mechanism design approach. *IEEE Transactions on Automatic Control*, 67(5), 2633–2639.
- Ekpo, P., Agarwal, S., Grimm, F., Molu, L., and Taylor, A. (2025a). Adafair-marl: Enforcing adaptive fairness constraints in multi-agent reinforcement learning. doi:10.48550/ARXIV.2511.14135. URL <https://arxiv.org/abs/2511.14135>.
- Ekpo, P.O., La, B., Wiener, T., Agarwal, S., Agrawal, A., Gonzalez-Pumariega, G., Molu, L., and Taylor, A. (2025b). Skill-aligned fairness in multi-agent learning for collaboration in healthcare. doi:10.48550/arXiv.2508.18708. URL <http://arxiv.org/abs/2508.18708>.
- Facchinei, F. and Kanzow, C. (2010). Generalized nash equilibrium problems. *Annals of Operations Research*, 175(1), 177–211.
- Jain, R., Chiu, D.M.W., and Hawe, W.R. (1984). A quantitative measure of fairness and discrimination for resource allocation in shared computer systems.
- Liang, S., Yi, P., and Hong, Y. (2017). Distributed nash equilibrium seeking for aggregative games with coupled constraints. *Automatica*, 85, 179–185.
- Neumann, J., Vogel, C., Kießling, L., Hempel, G., Kleber, C., Osterhoff, G., and Neumuth, T. (2024). Traumaflow—development of a workflow-based clinical decision support system for the management of severe trauma cases. *International Journal of Computer Assisted Radiology and Surgery*, 19(12), 2399–2409. Publisher: Springer.
- Nise, N.S. (2019). *Control Systems Engineering, Seventh Edition*. John Wiley & Sons.
- Rosenblum, S., Rab, S.L., and Admon, R. (2025). Dynamics in physiological acute stress response trajectories: uncovering latent variability. *BMC Psychiatry*, 25(1), 361. doi:10.1186/s12888-025-06807-2.
- Salinas-Avila, J.E., Gonzalez-Hernandez, H.G., Martinez-Chan, N., Bielma-Avendano, C.M., Salinas-Molar, X.A., and Garcia-Garcia, M.O. (2022). Assistant delivery robot for nursing home using ros: Robotic prototype for medicine delivery and vital signs registration. In *Proceedings of the 2022 5th International Conference on Electronics, Communications and Control Engineering*, 141–148.
- Sarcevic, A., Marsic, I., and Burd, R.S. (2012). Teamwork errors in trauma resuscitation. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 19(2), 1–30.
- Sun, L.W. and Yeung, C.K. (2007). Port placement and pose selection of the da vinci surgical system for collision-free intervention based on performance optimization. In *2007 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 1951–1956. IEEE.
- Tassa, Y., Erez, T., and Todorov, E. (2012). Synthesis and stabilization of complex behaviors through online trajectory optimization. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 4906–4913. doi:10.1109/IROS.2012.6386025.
- Taylor, A., Tanjim, T., Cao, H., and Lee, H.R. (2024). Towards collaborative crash cart robots that support clinical teamwork. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction, HRI '24*, 715–724. Association for Computing Machinery, New York, NY, USA.
- Weill Cornell Medicine (2025). Base Camp. URL <https://tinyurl.com/2t9musvp>.